

StaffSpire

Employee Management System

Objective:

Developing a Role-Based Employee Management System with Authentication, Authorization, Dashboard Analytics, Attendance Management, Leave Management, Reports, and Full CRUD Operations.

Being Developed By:

Softspire Solutions

Developer:

Mr. Shreyash Anawane (Full Stack Developer)

Entire Development Plan

Day 1: Project Setup

Frontend:

- Create React App / Vite
- Install Dependencies
- Setup Folder Structure
- Setup React Router
- Create Layouts

Backend:

- Setup Express Server
- Connect MongoDB Atlas
- Setup Environment Variables
- Create Base Folder Structure

Install

Frontend:

- `npm install react-router-dom axios`

Backend:

- `npm install express mongoose bcryptjs jsonwebtoken cors dotenv`
- `npm install nodemon --save-dev`

Goal:

1. Frontend running
2. Backend running
3. MySQL connected

Day 2: Authentication System

Backend:

- Create:
 1. Register API
 2. Login API
 3. JWT Token Generation
 4. Password Hashing

Frontend:

- Create:
 1. Login Page
 2. Register Page

Goal:

- User can login successfully
- JWT stored

Day 3: Role-Based Access Control

Create roles:

- Admin
- Manager
- Employee

Backend:

- Role Middleware
- Protected Routes

Frontend:

- Protected Routes
- Role Redirects

Examples:

- Admin → Admin Dashboard
- Manager → Manager Dashboard
- Employee → Employee Dashboard

Goal:

Users see only their own panel

Day 4: Dashboard Layout

Create:

- Sidebar
- Navbar
- Footer
- Dashboard Cards

Pages:

- Dashboard
- Employees
- Departments
- Attendance
- Leaves
- Tasks
- Reports

Goal:

- Full navigation working

Day 5: Department CRUD

Backend:

- Create Department
- Get Departments
- Update Department
- Delete Department

Frontend:

- Department Table
- Department Form

Goal:

- Full Department CRUD

Day 6: Employee CRUD

Backend:

- Add Employee
- Get Employees
- Update Employee
- Delete Employee

Frontend:

- Employee List
- Add Employee Form
- Edit Employee Form
- View Employee

Goal:

Full Employee CRUD

Day 7: Testing Day

Test:

- Authentication
- Departments
- Employees
- RBAC
- Navigation

Fix:

- Validation Issues
- API Errors
- UI Bugs

Goal

Week 1 fully stable

Day 8: Attendance Module

Backend:

- Mark Attendance
- Get Attendance
- Monthly Attendance

Frontend:

- Attendance Page
- Attendance Table

Goal

Attendance working

Day 9: Leave Management

Backend

- Apply Leave
- Approve Leave
- Reject Leave

Frontend

- Leave Application Form
- Leave History
- Leave Approval Page

Goal

Leave workflow complete

Day 10: Task Management

Backend:

- Create Task
- Assign Task
- Update Task Status
- Delete Task

Frontend:

- Task List
- Task Form
- Task Status Updates

Goal:

Task management complete

Day 11: Reports Module

Create:

- Employee Report
- Attendance Report
- Leave Report
- Task Report

Frontend

- Report Tables
- Filters

Goal

Reports visible

Day 12: Dashboard Analytics

Cards:

- Total Employees
- Departments
- Present Employees
- Pending Leaves
- Active Tasks

Charts:

- Attendance Chart
- Leave Chart
- Task Chart

Use:

- `npm install recharts`

Goal:

Dashboard looks complete

Day 13: Settings & Profile

Create:

- View Profile
- Edit Profile
- Change Password
- Logout

Goal

User account management complete

Day 14: Final Polish & Deployment

Testing

Check:

- Authentication
- CRUD
- Attendance
- Leave
- Tasks
- Reports
- Charts

Final Deliverables:

- JWT Authentication
- Role-Based Access Control
- Employee CRUD
- Department CRUD
- Attendance Management
- Leave Management
- Task Management
- Dashboard Analytics
- Reports
- Profile Management
- Responsive UI
- Deployment